Task 1 – CAPM [15%]

Assume the CAPM is valid. You are given the following information:

Stock	Risk free rate	Expected return on the market portfolio	Expected return on the stock	β	Price of the stock in period 0	Expected price of the stock in period 1
A	10%	20%	10%		100	
B	10%	20%		1.5		125
C	10%		20%	2.0	200	

The stocks pay no dividends next period. This is enough information to complete the table. Calculate all the missing data.

$$
\begin{aligned}
E[\tilde{r}] &= r_f + (E[r_m] - r_f)\beta \\
E[\tilde{r}] &= \frac{E[\tilde{P}_1] - P_0}{P_0} \\
\beta &= \frac{E[\tilde{r}] - r_f}{E[\tilde{r}_m] - r_f} \\
E[\tilde{r}_m] &= \frac{E[\tilde{r}] - r_f}{\beta} + r_f \\
E[\tilde{P}_1] &= P_0(1 + E[\tilde{r}]) \\
P_0 &= \frac{E[\tilde{P}_1]}{1 + E[\tilde{r}]}
\end{aligned}
$$

Based on these formulas you get the following table

r_f	$E[\tilde{r}_m]$	$E[\tilde{r}_1]$	β	P_0	$E[\tilde{P}_1]$
10%	20%	10%	0	100	110
10%	20%	25%	1.5	100	125
10%	15%	20%	2.0	200	240

Task 2 – International Corporate Finance [20%]

Consider a Norwegian investor who is planning to invest in a South African risk-free security that pays out 480 RAND in one year. Assume that the risk free interest rate in South Africa is 4.6%. The current spot exchange rate of Norwegian Krone to South African Rand is 1.66 RAND/NOK

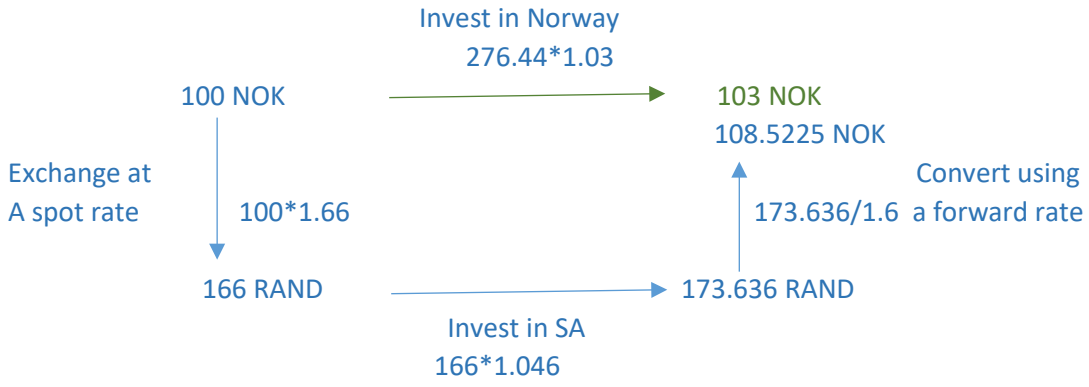
a) What will the investor pay for this security in NOK?

$$PV_{RAND} = \frac{C_{RAND}}{1 + r_{RAND}} = \frac{480}{1 + 0.046} = 458.89\ RAND$$

Price in NOK is 458.89/1.66=276.44.

Assume that the risk free interest rate in Norway is 3% and one-year forward rate is 1.6 RAND/NOK.

b) Is it possible for the investor Norwegian investor to profit from an arbitrage opportunity? If yes, illustrate this opportunity and calculate the arbitrage profit. If no, show that the arbitrage profit is zero. Start the illustration by assuming that an investor borrows 100 NOK.
Yes, it can, for example:



Arbitrage strategy is to borrow 100 NOK at a risk free rate in Norway, exchange to RAND at a spot rate and invest in South Africa and convert using the forward rate. This secures 108.5225 NOK as opposed to 103 that one would own in Norway. Profit is 108.5225-103=5.5225.

c) Under the assumptions in this task we can conclude that these markets are not internationally integrated. Why is that?
Interest rate parity does not hold

d) Find the forward rate that would be implied by the assumption of internationally integrated capital markets.
1.685 RAND/NOK or 0.59 NOK/RAND

Task 3 – Capital Structure [20%]

ABC Inc. is an all-equity firm with 10 million shares outstanding which trade at 15 NOK per share, and if faces a 40% corporate tax rate. Although investors currently expect ABC to remain an all-equity firm, ABC plans to announce tomorrow that it will issue perpetual debt for an amount of NOK 80 million and will use all the funds to repurchase shares. Market participants understand that with this new level of debt ABC could become financially distressed some time in the future. They also understand that the present value of the expected financial distress costs borne by shareholders is NOK 40 million. The announcement will also convey that ABC has no further plans to change its debt level.

a) What are the market value of ABC's assets before the announcement?
Because the firm is all-equity the value of it's assets is the value of its equity before the announcement $V_U = E = 10\ mil \cdot NOK15 = NOK150\ mil$

b) What is the market value of ABC's assets after debt is issued and the shares are repurchased? What is the market value of ABC's equity?

$$V_L = E_L + D = V_U + tD - PV(Distress\ costs) = 150\ mil + 0.4 \cdot 80mil - 40\ mil$$
$$= NOK\ 142\ mil$$

The value of equity

$$E_L = 142\ mil - 80\ mil = NOK\ 62\ mil.$$

c) What is the market value of ABC's equity and the share price upon announcement (but before execution) of the transaction?

Upon announcement the value of the equity decreases to NOK 142 mil and the share price drops to 14.2

d) How many shares will ABC repurchase?

Shares repurchased = $\frac{80\ mil}{14.20} = 5\ 633\ 803$

e) Is this change in capital structure beneficial to shareholders? Why?

The change in capital structure decreases firm value: the benefit (gain in interest tax shield) is less than the cost (cost of financial distress). Consequently, the debt issue takes the firms to the situation where the amount of debt exceeds the optimal amount that balances the marginal benefits and costs.

Task 4 – Option Valuation [30%]

Consider a stock with a current price of 400 NOK. In each of the next two years (2019 and 2020) the stock price will either increase by 25% or decrease by 20%. The stock pays no dividends and the risk free interest rate is equal to 7%. Use a two-period binomial model to answer the following questions

f) Calculate the price of a European call option on this stock with a strike price of 375 NOK.

Assume now that the stock pays a 25% dividend in period 2019.

g) Recalculate the price of the European option in a) given new assumption regarding dividends. (***hint:*** you need to calculate the option values from 2019 onwards based on ex dividend price, whereas the option value in 2018 based on the cum dividend price)

h) Compare the prices in a) and b) and explain the difference (or the absence of difference).

i) What is the price of the American option on the same stock as in b) with the same strike price of 375 NOK? (the same ***hint*** as in b) applies).

j) Compare the prices of American and European option in b) and d) and explain the difference (or the absence of difference).

Note: you can choose between replicating portfolio method or risk-neutral probabilities to price European options.

a) Binomial tree and values(in parentheses)

2018	2019	2020
		625
		(250)
	500	
400	(149.53)	400
(89.09)		(25)

320

(14.02)

256

(0)

Replicating portfolio: Δ S+B

In 2019: up state Δ=1 and B=-350.47; down state Δ=0.1736 and B=-41.53.

In 2018: Δ=0.7528 and B=-212.03.

Alternatively, one could use risk-neutral probabilities. Pu=(r-d)/(u-d)=(1.07-0.8)/(1.25-0.8)=0.6; Pd=0.4. Option value in 2018: (0.6*0.6*250+2*0.6*0.4*24)/(1.07)^2=89.09

b) Binomial tree and values(in parentheses) for the European Option

2018	2019		2020
			468.75
	(cum div)	500	(93.75)
	(ex div)	375	
400	(52.57)		300
(29.46)			(0)
	(cum div)	320	
	(ex div)	240	
	(0)		192
			(0)

Replicating portfolio: Δ S+B

In 2019: up state Δ=0.5556 and B=-155.7757.

In 2018: Δ=0.2921 and B=-87.357.

Alternatively, one could use risk-neutral probabilities. Pu=(r-d)/(u-d)=(1.07-0.8)/(1.25-0.8)=0.6; Pd=0.4. Option value in 2018: (0.6*0.6*93.75)/(1.07)^2=29.47.

c) The price is b is much lower, since the option became less valuable in the presence of dividends.

d) Binomial tree and values(in parentheses) for the American Option

2018	2019		2020
			468.75
	(cum div)	500	(93.75)
	(ex div)	375	
400	**(125)**		300
(29.46)			(0)
	(cum div)	320	
	(ex div)	240	
	(0)		192
			(0)

In 2019: payoff is max(52.57, 500-375)= 125

In 2018: Δ=0.2921 and B=-87.357.

Or 0.6*125/1.07=70.09

e) The price of the American option is higher as there is due to the value of early exercise: it is valuable to exercise the option in 2019.

Task 5 – Law of One Price [15%]

In the discussions of some of the models in this course, we relied on the no-arbitrage argument or the Law of One Price: "If two investment strategies have the same payoff tomorrow, they must have the same price today."

For which of the following theories/relationships, a) shortly explaining each theory/relationship and b) explain which of the following theories relies explicitly on no-arbitrage arguments.

i. Relationship between the value of coupon bonds using zero-coupon bond. Use an example to illustrate the answer to this question (you need to make some assumptions about coupon payments, time to maturity and face value).

Yes, by law of one price if using the zero-coupon bonds you can replicate the same cash flows as coupon the cash flows as risk are the same. Consider the following example.

	0	1	2	3
Coupon bond:		$100	$100	$1100
1-year zero:		$100		
2-year zero:			$100	
3-year zero:				$1100
Zero-coupon Bond portfolio:		$100	$100	$1100

Maturity	1 year	2 years	3 years	4 years
YTM	3.50%	4.00%	4.50%	4.75%
Price	$96.62	$92.45	$87.63	$83.06

Zero-Coupon Bond	Face Value Required	Cost
1 year	100	96.62
2 years	100	92.45
3 years	1100	11 × 87.63 = 963.93
	Total Cost:	$1153.00

By the Law of One Price, year coupon bond must trade for a price of $1153.

ii. The Valuation of Stocks based on Multiples

The multiples approach is a valuation theory based on the idea that similar assets sell at similar prices. This assumes that a ratio comparing value to some firm-specific variable (operating margins, cash flow, etc.) is the same across similar firms.
Direct implication of the Law of one price effect. If we replicate same cashflows with the same risk we pay the same price. Estimate the value of the firm based on the value of other, comparable firms or investments that we expect will generate very similar cash flows in the future.

iii. Fama-French-Carhart multifactor model of risk

Fama-French-Carhart Factor Specification

$$E[R_s] = r_f + \beta_s^{Mkt}(E[R_{Mkt}] - r_f) + \beta_s^{SMB}E[R_{SMB}] + \beta_s^{HML}E[R_{HML}] + \beta_s^{PR1YR}E[R_{PR1YR}]$$

In order to estimate cost of capital it is not a necessary to identify the efficient portfolio itself. As long as we can identify a collection of well-diversified portfolios from which an efficient portfolio can be constructed, we can use it to measure risk. Thus, we can write the risk premium of any marketable security as the sum of the risk premium of each factor multiplied by the sensitivity of the stock with that factor—the factor betas.

This theory similarly to CAPM does not explicitly rely on no-arbitrage arguments.